



Applying role-based security

Security and Policies



Security Overview

Two levels of security

- **Platform-level security** defines access to features in the platform.
E.g.: data location, logging configuration, and plug-ins administration
- **Model level security** defines access to data in the models.
E.g.: entity access, rows and column access



Security is role-based

- **Privileges** are granted to **Roles** defined in Semarchy xDM
- **Roles** are associated to **Users**

How does security work? An example

1. User John authenticates with his *john.doe* SSO account to the Semarchy application
 - Semarchy establishes the connection to the enterprise **Authentication Server** with the **Identity Provider**.
2. The session gets a context with a list of **Roles**
 - Those roles have matches in Semarchy and associated platform-level and model-level **privileges** are granted
3. The Semarchy application opens
 - Semarchy will filter the welcome page UI content and only allow operations based on the **privileges granted**
 - The same mechanism applies when using Semarchy web services

Roles and Users

Two pre-defined **system roles** exist in Semarchy:

- **semarchyAdmin**: Role with full privileges
- **semarchyConnect**: Role to connect to the Semarchy application. **Required for all Semarchy users**

Customized Roles allow specific privilege grants:

- They are created by the Semarchy administrator in the Semarchy configuration console



Privileges Precedence

Three types of privileges exist

- **Read/Write, Read** and **None**

Privileges apply in order of precedence

- Read/Write **then** Read **then** None
- E.g.: *JohnDoe* has two profiles given to him:
 - *Developer* has **Read** privileges on **model design**
 - *Operator* has **None** privileges on **model design**
 - The resulting privilege for the user is **Read** on **model design**



A user always has the best privilege associated to his roles



Securing data access

Roles



Roles Management

- Configured from **Configuration > Roles**
- Internally and externally assigned:
 - Roles lookup with a table database
 - Identity Provider role mapping

 **Data Steward** 

GENERAL

NameDataSteward

LabelData Steward

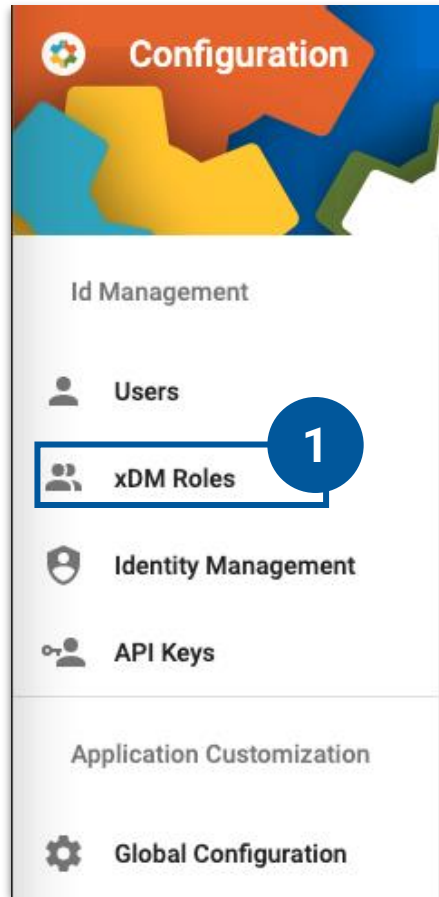
Emailsdata.stewards@semarchy.com

DescriptionGroup of data stewards inside Semarchy Organization.

Privileges

	None	Read	Read/Write
Application Design	☒	☐	☐
Application Management	☒	☐	☐
Dashboard Design	☐	☒	☐
Dashboard Management	☒		☐
Discovery Management	☒		☐
Platform Admin	☒	☐	☐
Role Management	☒		☐
User Management	☒		☐

Creating Roles (Console Perspective)



Roles			
☐	Name	Label	Err
☐	DataSteward	Data Steward	
☐	BusinessUser	Business User	
☐	semarchyAdmin		
☐	semarchyConnect		

Add Role

Name*

Marketing

Label

Marketing

Privileges

	None	Read	Read/Write
Application Design	☒	☐	☐
Application Management	☒	☐	☐
Dashboard Design	☒		☐
Dashboard Management	☒		☐
Discovery Management	☒		☐
Platform Admin	☒	☐	☐
Role Management	☒		☐
User Management	☒		☐

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CREATE



Securing data access

Users



Users Management

- Configured from **Configuration > Users**
- Replaces tomcat-users and DB Realm
- SSO users automatically created at first login
- Support Roles assignment
- Support password management
- Users import/export from the UI and REST API

Configuration > Users					
Users					
☐	Username	Assigned Roles	First Name	Last Name	Email
	accountmanager	Account Manager, semarchyConnect	Aciva	Mason	info@semarchy.com
	businessuser	Business User, semarchyConnect	Buster	Usuah	bu@semarchy.com
	datasteward	Data Steward, semarchyConnect	Dana	Steuart	ds@semarchy.com

Identity Management

- Built-in configuration & provides a tool for testing
- Internal Identity Provider
- SSO compatible with the following providers:
 - OpenID Connect
 - SAML
 - LDAP
 - Active Directory
 - Windows Authentication (SSO and Form)
- Provides interface to manager Roles & Profiles
- Allows configuration of Login page

Google (OpenID Connect)

CONFIGURATION ROLES MAPPING PROFILE BUTTON

General

Name*
Google

Provider type*
OpenID Connect

Latest Connection
Failed - Apr 7, 2021 4:49:03 pm

Configuration Properties

Connectivity

Issuer Identifier*
https://accounts.google.com

Client ID*
[Redacted]

Client Secret*
[Redacted]

Redirect URL
/login/oauth2/code/Google

Logout URL
https://accounts.google.com/logout

User Authentication

Additional Scopes
email

Username Claim*
email

Login Page

Semarchy xDM

LOGIN WITH GOOGLE

OR

User name

Password

LOG IN



Applying role-based security

Platform privileges



Platform Features

- **Application Design:** Grants access to all the components of the Model Design perspective
- **Application Management:** Grants access to all components for model and application management
- **Dashboard Design:** Grants limited access to the Dashboard Builder to create charts and dashboards only
- **Dashboard Management:** Grants access to the Dashboard Builder
- **Discovery Management:** Grants access to xDM Discovery
- **Platform Admin:** Grants access to the Semarchy Configuration interface
- **Role Management:** Grants access to the xDM roles configuration
- **User Management:** Grants access to the User configuration

 **Data Steward** 

GENERAL

Name

DataSteward

Label

Data Steward

Emails

Description

Privileges

	None	Read	Read/Write
Application Design	☒	☐	☐
Application Management	☒	☐	☐
Dashboard Design	☒		☐
Dashboard Management	☒		☐
Discovery Management	☒		☐
Platform Admin	☒	☐	☐
Role Management	☒		☐
User Management	☒		☐

Restricted platform features

The following platform features cannot be handled by customized roles

They are only allowed to users with the system **semarchyAdmin** role:

- Notification Servers
- Identity Management
- Repository Setup
- License Changes





Securing data access

Model level security



Required Role

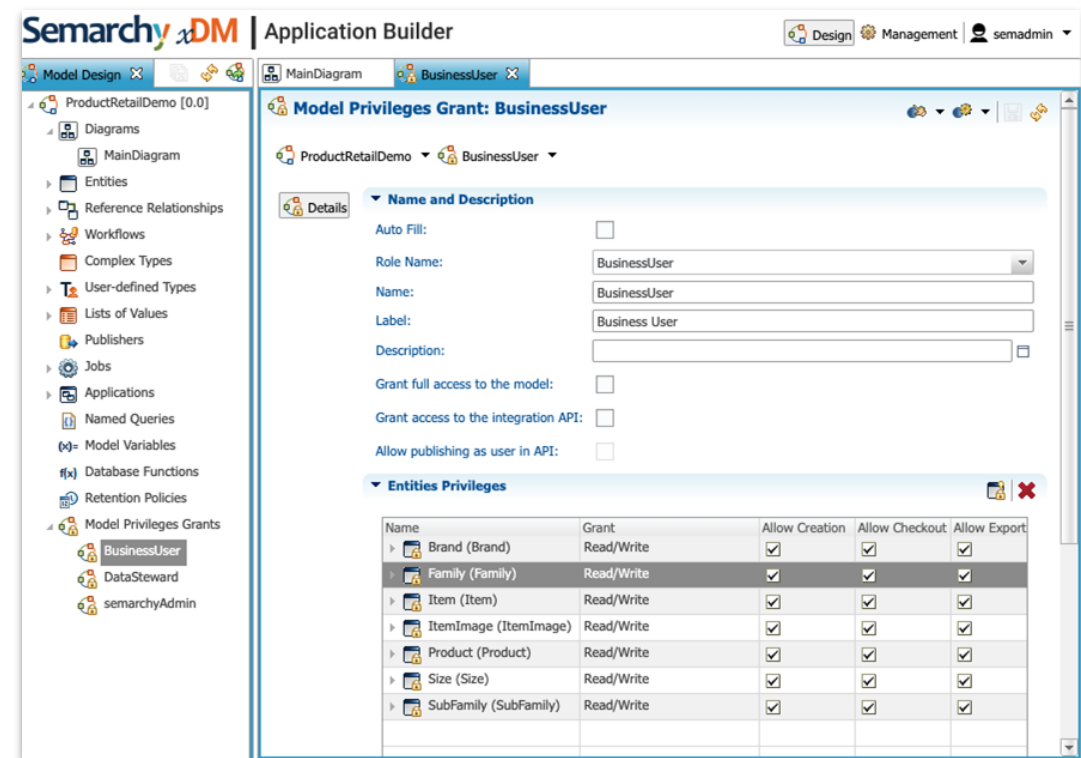
The following components can require a role to be available

Application access	Dashboard
Browsing history	Business view
Chart	Action
Workflow (admin, task & transition)	

Roles

Roles directly impact the actions and features available:

- Entities/Attributes with no read privileges do not appear
- Entities/Attributes with no write privileges cannot be modified
- Records cannot be added from entities without the create privileges
- Exporting data (Excel) is not possible without export privileges



Privilege Types

Model Privileges Grants at **Entity level**: *None, Read, Read/Write, Create, Checkout, Export, Remove and Delete*

Model Privileges Grants at **Attribute level**: *None, Read, Read/Write*

- E.g.: All users can read the *full name* and *title* attributes from the *employee* entity. Only users with the *HRManager* role can read sensitive attributes (*private email, SSID and salary*)

For the **Privileges Types** see

https://www.semarchy.com/doc/semarchy-xdm/xdm/5.3/Design/security/data-hub-security.html#_privileges_types

Create New Entities Privileges Grants

Entity Privileges Grant
Configure Default Privileges

Default Grant: None

Allow Creation: ☐

Allow Checkout: ☐

Allow Export: ☐

Allow Removal: ☐

Allow Delete: ☐

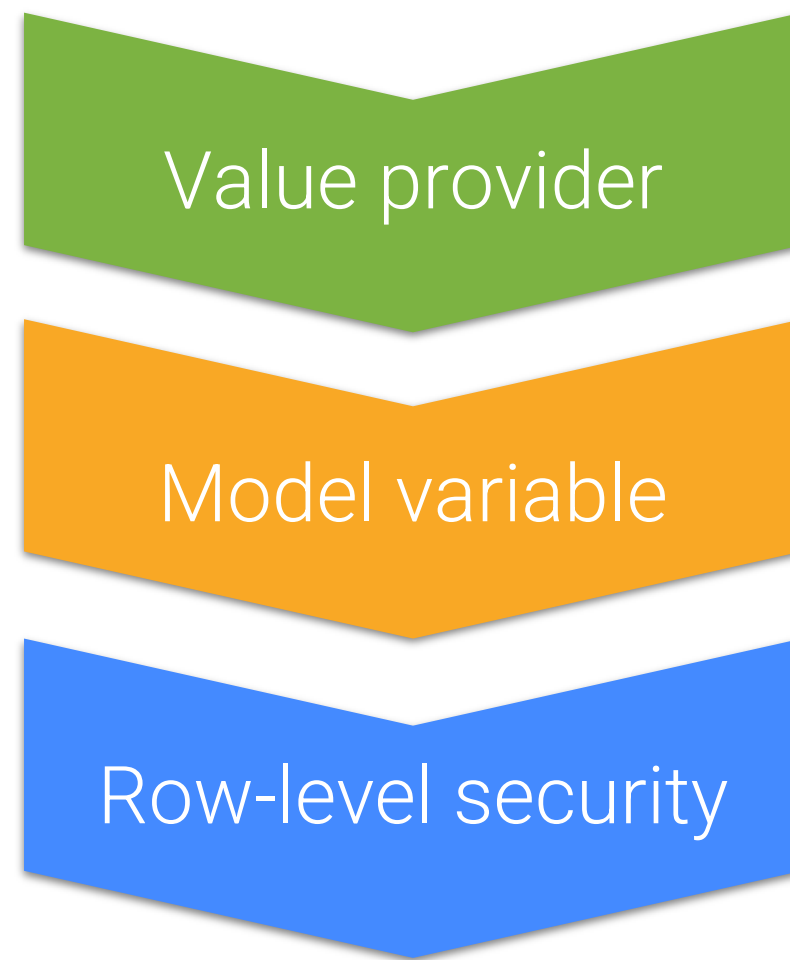
< Back Next > Finish Cancel

Row Level Security

Privileges may apply to a subset of the rows in the context of a user session.

The subset is filtered by a SemQL expression using:

- **User session variables**
(e.g.: V_USER_COUNTRY)
- **Model variables:**
 - Refreshed from **variable value providers**
 - Possibly derived from session variables (e.g.: Retrieve from LDAP the user's region using his country)



Row Level Security

Entity Privileges can apply a subset of the rows using **Row-level Filtering**. This security is contextual to the user session.

- E.g.: Users can only see the *Customers* which country correspond to their own country defined in the LDAP directory

BusinessUser X

Model Privileges Grant: BusinessUser

CustomerTraining BusinessUser

Details

Name and Description

Entities Privileges

Name	Grant	Allow Creation	Allow Checkout	Allow Export	Allow Removal	Allow Delete	Filter
BankDetail (BankDetail)	Read	☐	☐	☐	☐	☐	
Customer (Customer)	Read	☒	☒	☒	☒	☒	Address.Country = :V_USER_COUNTRY

Data access control example

